DLNM-LC-LL Calibration

Load Packages

```
packages <- c(
  "readxl", "ggplot2", "forecast", "dplyr", "writexl", "tseries",
  "hwwntest", "dlnm", "splines", "ISOweek", "mgcv", "Metrics",
  "demography", "seastests", "MTS", "reshape2", "astsa", "MASS",
  "patchwork", "scales", "tidyr"
)
invisible(lapply(packages, library, character.only = TRUE))
```

Project Root

```
find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) {
      return(path)
    }
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()

combined_dir <- file.path(base_dir, "Data/Combined_data")
function_dir <- file.path(base_dir, "Code/Function")
figure_calibration_dir <- file.path(base_dir, "Figures/Calibration")
figure_overall_dir <- file.path(base_dir, "Figures/DLNM/Overall")
results_dir <- file.path(base_dir, "Results/Calibration")

dir.create(figure_calibration_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(figure_overall_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(results_dir, recursive = TRUE, showWarnings = FALSE)
```

Load Data

```
files <- list.files(function_dir, pattern = "\\.R$", full.names = TRUE)
invisible(sapply(files, source))

age_names <- c("Y20_64", "Y65_74", "Y75_84", "Y_GE85")
```

```r
age_labels <- c(Y20_64 = "20--64", Y65_74 = "65--74",
                Y75_84 = "75--84", Y_GE85 = "85+")
age_titles <- c(Y20_64 = "Age 20-64", Y65_74 = "Age 65-74",
                Y75_84 = "Age 75-84", Y_GE85 = "Age 85+")
regions <- c("Athens", "Lisbon", "Rome")

Y20_64 <- read_xlsx(file.path(combined_dir, "Y20_64_combined.xlsx"))
Y65_74 <- read_xlsx(file.path(combined_dir, "Y65_74_combined.xlsx"))
Y75_84 <- read_xlsx(file.path(combined_dir, "Y75_84_combined.xlsx"))
Y_GE85 <- read_xlsx(file.path(combined_dir, "Y_GE85_combined.xlsx"))

Athens <- cbind(Y20_64[[3]] / Y20_64[[9]], Y65_74[[3]] / Y65_74[[9]],
                Y75_84[[3]] / Y75_84[[9]], Y_GE85[[3]] / Y_GE85[[9]]) * 52
Lisbon <- cbind(Y20_64[[4]] / Y20_64[[10]], Y65_74[[4]] / Y65_74[[10]],
                Y75_84[[4]] / Y75_84[[10]], Y_GE85[[4]] / Y_GE85[[10]]) * 52
Rome <- cbind(Y20_64[[6]] / Y20_64[[12]], Y65_74[[6]] / Y65_74[[12]],
              Y75_84[[6]] / Y75_84[[12]], Y_GE85[[6]] / Y_GE85[[12]]) * 52
colnames(Athens) <- colnames(Lisbon) <- colnames(Rome) <- age_names

Athens_wave <- data.frame(Athens_hot_wave3 = Y20_64$Attiki_hot_wave3,
                          Athens_cold_wave3 = Y20_64$Attiki_cold_wave3)
Lisbon_wave <- data.frame(Lisbon_hot_wave3 = Y20_64$Lisbon_hot_wave3,
                          Lisbon_cold_wave3 = Y20_64$Lisbon_cold_wave3)
Rome_wave <- data.frame(Rome_hot_wave3 = Y20_64$Roma_hot_wave3,
                        Rome_cold_wave3 = Y20_64$Roma_cold_wave3)

dat_list <- list(Athens = Athens, Lisbon = Lisbon, Rome = Rome)
wave_list <- list(Athens = Athens_wave, Lisbon = Lisbon_wave, Rome = Rome_wave)
```

Cross-Basis Matrices

```r
build_crossbasis_matrices(base_dir)

make_cb <- function(utci_ext) {
  varknots <- equalknots(utci_ext, fun = "ns", df = 3, degree = 3)
  lagknots <- logknots(21, fun = "ns", df = 3)
  crossbasis(
    utci_ext,
    lag = 21,
    argvar = list(fun = "ns", knots = varknots),
    arglag = list(knots = lagknots, df = 3)
  )
}

UTCI_ext.Athens <- load_crossbasis_history(base_dir, "Athens", "Attiki")
UTCI_ext.Lisbon <- load_crossbasis_history(base_dir, "Lisbon", "Lisbon")
UTCI_ext.Rome <- load_crossbasis_history(base_dir, "Rome", "Roma")

Athens_cb <- make_cb(UTCI_ext.Athens)
Lisbon_cb <- make_cb(UTCI_ext.Lisbon)
Rome_cb <- make_cb(UTCI_ext.Rome)
cb_list <- list(Athens = Athens_cb, Lisbon = Lisbon_cb, Rome = Rome_cb)
```

Fit DLNM–LC And DLNM–LL

```
invisible(capture.output({
  b <- DLNM_LL(dat_list = dat_list, wave_list = wave_list,
                cb_list = cb_list, tol = 1e-2, max_iter = 20)
  b1 <- DLNM_LC(Athens, Athens_cb, wave_data = Athens_wave,
                 tol = 1e-2, max_iter = 20, region = "Athens")
  b2 <- DLNM_LC(Lisbon, Lisbon_cb, wave_data = Lisbon_wave,
                 tol = 1e-2, max_iter = 20, region = "Lisbon")
  b3 <- DLNM_LC(Rome, Rome_cb, wave_data = Rome_wave,
                 tol = 1e-2, max_iter = 20, region = "Rome")
}))

saveRDS(list(b = b, b1 = b1, b2 = b2, b3 = b3),
        file.path(results_dir, "DLNM_LC_LL_calibration_models.rds"))
```

Figure 3. Fitted time-varying factors in LC and DLNM–LC

```
DLNM_LC_models <- list(Athens = b1, Lisbon = b2, Rome = b3)

plot_lc_kappa <- function(region_name, model, model_type, y_limits) {
  df <- data.frame(
    time = seq(as.Date("2015-01-01"), as.Date("2019-12-30"), length.out = 260),
    kappa_t = if (model_type == "LC") {
      as.numeric(model$baseline_LC$k_t)
    } else {
      as.numeric(model$final_LC$k_t)
    },
    Model = model_type
  )

  ggplot(df, aes(x = time, y = kappa_t, color = Model)) +
    geom_line(size = 0.5) +
    scale_color_manual(name = "",
                       values = if (model_type == "LC") c("LC" = "black")
                       else c("DLNM-LC" = "darkred")) +
    scale_x_date(limits = as.Date(c("2015-01-01", "2020-01-01")),
                 date_breaks = "1 year", date_labels = "%Y", expand = c(0, 0.01)) +
    scale_y_continuous(limits = y_limits) +
    labs(title = bquote(bold(.(region_name))), x = "Year",
         y = expression(hat(kappa)(t))) +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold", size = 18),
      axis.title = element_text(size = 16),
      axis.text = element_text(size = 14),
      legend.text = element_text(size = 14),
      panel.grid = element_blank(),
      panel.border = element_rect(color = "black", fill = NA, size = 1),
      legend.position = "none"
    )
}
```

```r
all_kappa_values <- unlist(lapply(DLNM_LC_models, function(model) {
  c(model$baseline_LC$k_t, model$final_LC$k_t)
}))
y_limits <- range(all_kappa_values, na.rm = TRUE)

LC_plots <- (
  plot_lc_kappa("Athens", b1, "LC", y_limits) /
    plot_lc_kappa("Lisbon", b2, "LC", y_limits) /
    plot_lc_kappa("Rome", b3, "LC", y_limits)
) + plot_layout(guides = "collect") & theme(legend.position = "bottom")

DLNM_LC_plots <- (
  plot_lc_kappa("Athens", b1, "DLNM-LC", y_limits) /
    plot_lc_kappa("Lisbon", b2, "DLNM-LC", y_limits) /
    plot_lc_kappa("Rome", b3, "DLNM-LC", y_limits)
) + plot_layout(guides = "collect") & theme(legend.position = "bottom")

single_model_k_plot <- (LC_plots | DLNM_LC_plots) &
  theme(plot.margin = grid::unit(c(5, 20, 5, 5), "pt"))

ggsave(file.path(figure_calibration_dir, "single_model_k.pdf"),
       single_model_k_plot, width = 12, height = 6, dpi = 300)
print(single_model_k_plot)
```

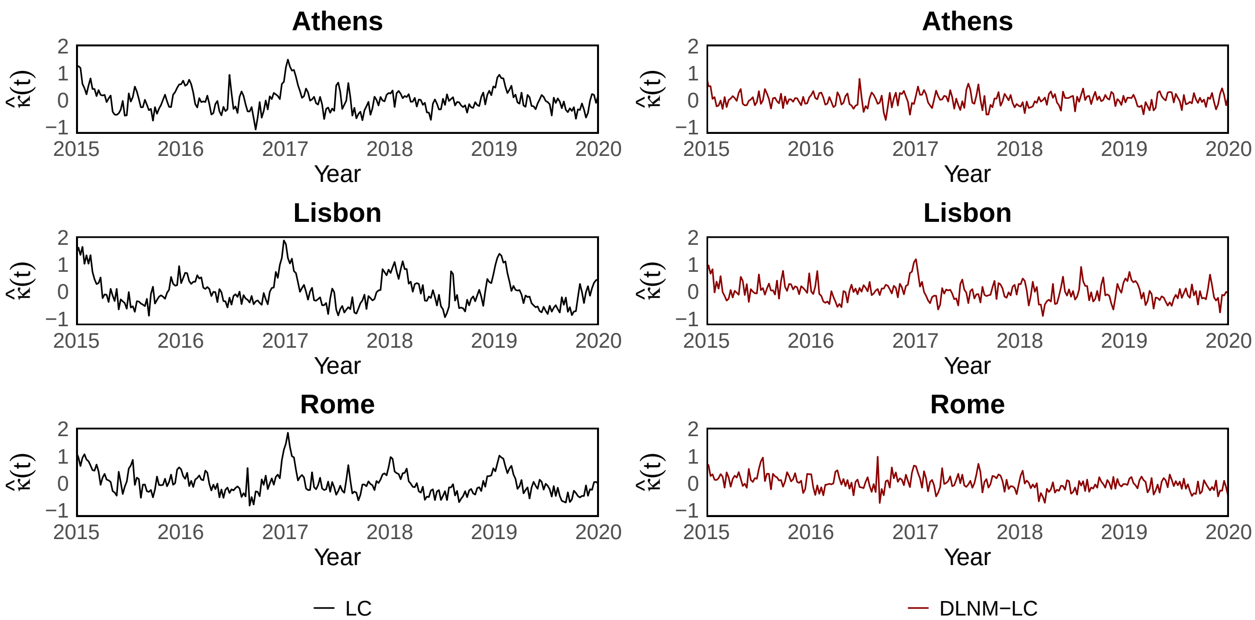
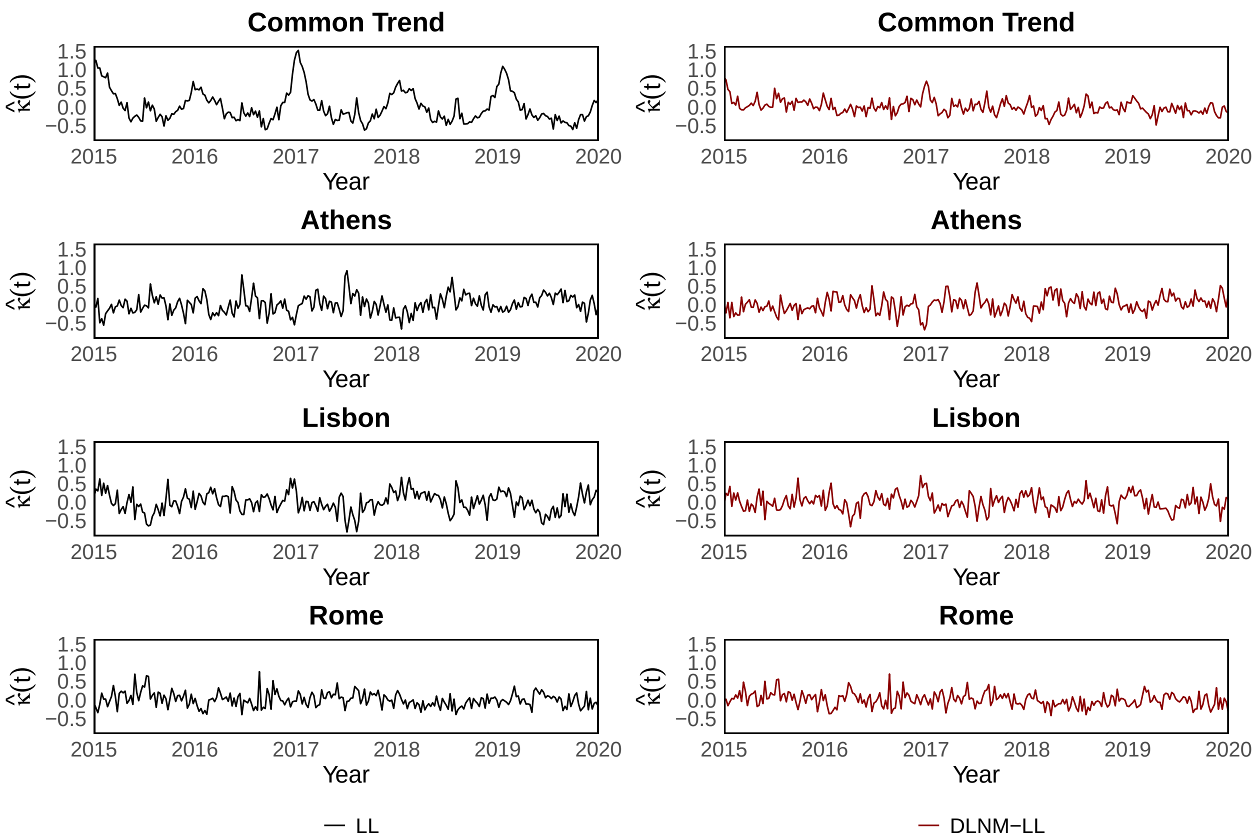


Figure 4. Fitted time-varying factors in LL and DLNM–LL

```r
plot_ll_kappa <- function(region_name, base_kt, final_kt, model_type, y_limits) {
  df <- data.frame(
    time = seq(as.Date("2015-01-01"), as.Date("2019-12-30"), length.out = 260),
    kappa_t = if (model_type == "LL") as.numeric(base_kt) else as.numeric(final_kt),
    Model = model_type
  )
```

```r
  ggplot(df, aes(x = time, y = kappa_t, color = Model)) +
    geom_line(size = 0.5) +
    scale_color_manual(name = "",
                       values = if (model_type == "LL") c("LL" = "black")
                       else c("DLNM-LL" = "darkred")) +
    scale_x_date(limits = as.Date(c("2015-01-01", "2020-01-01")),
                 date_breaks = "1 year", date_labels = "%Y", expand = c(0, 0.01)) +
    scale_y_continuous(limits = y_limits) +
    labs(title = bquote(bold(.(region_name))), x = "Year",
         y = expression(hat(kappa)(t))) +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold", size = 18),
      axis.title = element_text(size = 16),
      axis.text = element_text(size = 14),
      legend.text = element_text(size = 14),
      panel.grid = element_blank(),
      panel.border = element_rect(color = "black", fill = NA, size = 1),
      legend.position = "none"
    )
}

all_kappa_values <- c(
  b$baseline_LL$Kt, b$final_LL$Kt,
  b$baseline_LL$kt$Athens, b$final_LL$kt$Athens,
  b$baseline_LL$kt$Lisbon, b$final_LL$kt$Lisbon,
  b$baseline_LL$kt$Rome, b$final_LL$kt$Rome
)
y_limits <- range(all_kappa_values, na.rm = TRUE)

LL_plots <- (
  plot_ll_kappa("Common Trend", b$baseline_LL$Kt, b$final_LL$Kt, "LL", y_limits) /
    plot_ll_kappa("Athens", b$baseline_LL$kt$Athens, b$final_LL$kt$Athens, "LL", y_limits) /
    plot_ll_kappa("Lisbon", b$baseline_LL$kt$Lisbon, b$final_LL$kt$Lisbon, "LL", y_limits) /
    plot_ll_kappa("Rome", b$baseline_LL$kt$Rome, b$final_LL$kt$Rome, "LL", y_limits)
) + plot_layout(guides = "collect") & theme(legend.position = "bottom")

DLNM_LL_plots <- (
  plot_ll_kappa("Common Trend", b$baseline_LL$Kt, b$final_LL$Kt, "DLNM-LL", y_limits) /
    plot_ll_kappa("Athens", b$baseline_LL$kt$Athens, b$final_LL$kt$Athens, "DLNM-LL", y_limits) /
    plot_ll_kappa("Lisbon", b$baseline_LL$kt$Lisbon, b$final_LL$kt$Lisbon, "DLNM-LL", y_limits) /
    plot_ll_kappa("Rome", b$baseline_LL$kt$Rome, b$final_LL$kt$Rome, "DLNM-LL", y_limits)
) + plot_layout(guides = "collect") & theme(legend.position = "bottom")

multi_model_k_plot <- (LL_plots | DLNM_LL_plots) &
  theme(plot.margin = grid::unit(c(5, 20, 5, 5), "pt"))

ggsave(file.path(figure_calibration_dir, "multi_model_k.pdf"),
       multi_model_k_plot, width = 12, height = 8, dpi = 300)
print(multi_model_k_plot)
```



DLNM Climate-Driven Components

```r
LC_models <- list(Athens = b1, Lisbon = b2, Rome = b3)
LC_temp <- LL_temp <- theta_LC <- theta_LL <- list()

for (region in regions) {
  LC_model <- LC_models[[region]]
  LC_a <- as.data.frame(matrix(rep(LC_model$final_LC$a_x, times = 260),
                               ncol = 4, byrow = TRUE))
  LL_a <- as.data.frame(matrix(rep(b$final_LL$Ax[[region]], times = 260),
                               ncol = 4, byrow = TRUE))

  LC_temp[[region]] <- as.data.frame(
    log(as.matrix(dat_list[[region]])) - LC_model$final_LC$log_fitted
  )
  LL_temp[[region]] <- as.data.frame(
    log(as.matrix(dat_list[[region]])) - b$final_LL$log_fitted[[region]]
  )
  colnames(LC_temp[[region]]) <- colnames(LL_temp[[region]]) <- age_names

  LC_ratio <- exp(LC_model$log_fitted - LC_a - LC_model$final_LC$std_mxt)
  LL_ratio <- exp(b$log_fitted[[region]] - LL_a - b$final_LL$std_mxti[[region]])
  colnames(LC_ratio) <- colnames(LL_ratio) <- age_names

  theta_LC[[region]] <- 1 - 1 / LC_ratio
  theta_LL[[region]] <- 1 - 1 / LL_ratio
```

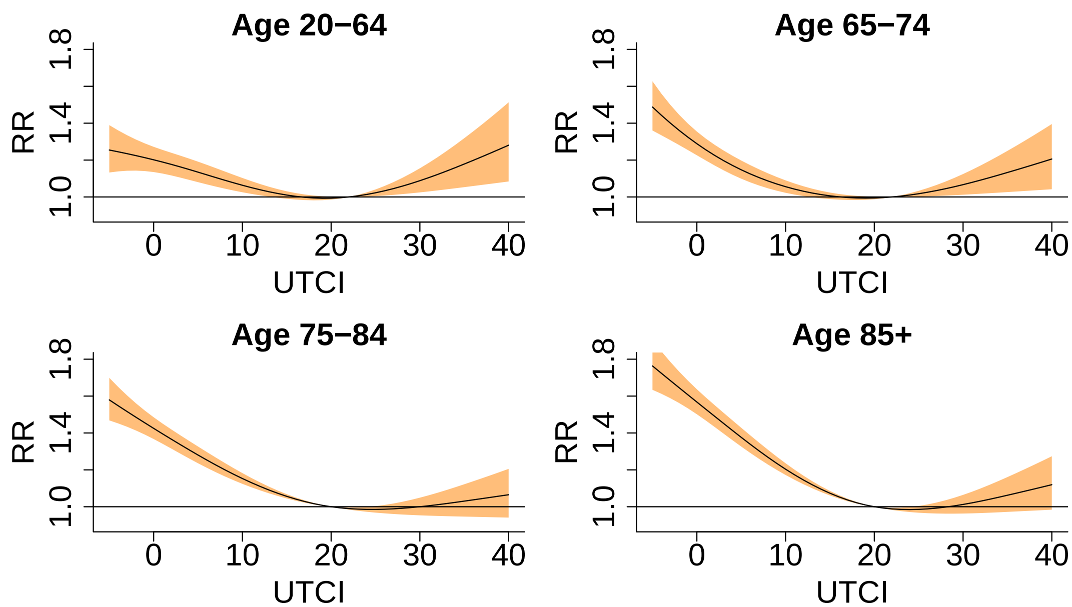
```
}
```

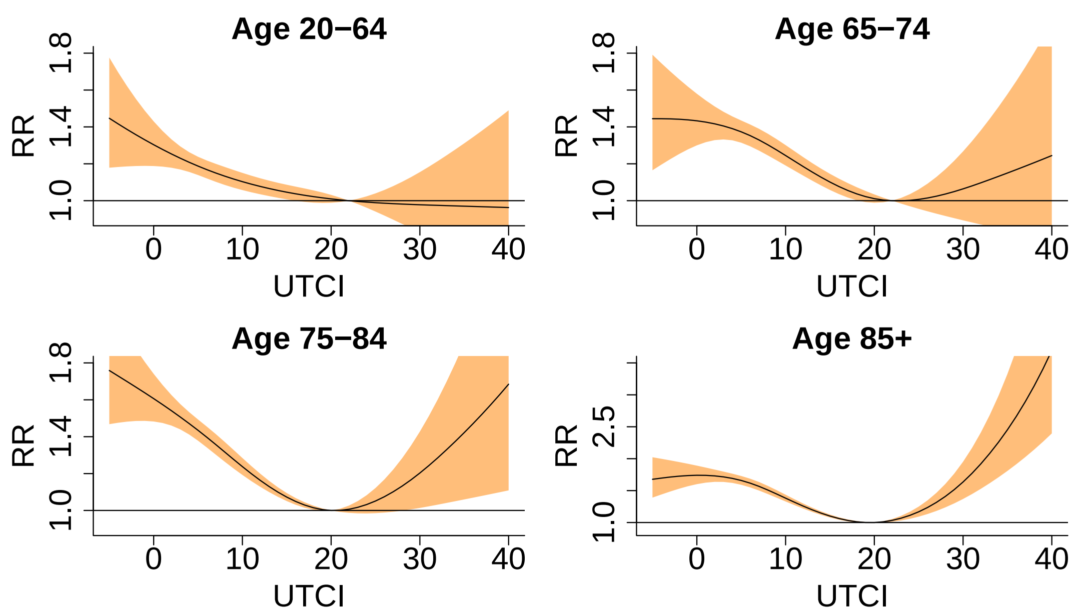
Figure 5. Overall cumulative UTCI effects

```
overall_centers <- list(
  Athens = c(Y20_64 = 22, Y65_74 = 22, Y75_84 = 20, Y_GE85 = 20),
  Lisbon = c(Y20_64 = 22, Y65_74 = 22, Y75_84 = 20, Y_GE85 = 20),
  Rome = c(Y20_64 = 18, Y65_74 = 20, Y75_84 = 20, Y_GE85 = 20)
)

overall_ylim <- list(
  Athens = list(Y20_64 = c(0.9, 1.8), Y65_74 = c(0.9, 1.8),
                Y75_84 = c(0.9, 1.8), Y_GE85 = c(0.9, 1.8)),
  Lisbon = list(Y20_64 = c(0.9, 1.8), Y65_74 = c(0.9, 1.8),
                Y75_84 = c(0.9, 1.8), Y_GE85 = c(0.9, 3.5)),
  Rome = list(Y20_64 = c(0.9, 1.8), Y65_74 = c(0.9, 1.8),
              Y75_84 = c(0.9, 1.8), Y_GE85 = c(0.9, 1.8))
)

fit_overall_pred <- function(region, age_group) {
  cb <- cb_list[[region]]
  wave_data <- wave_list[[region]]
  y <- LL_temp[[region]][[age_group]]
  fit <- glm(
    y ~ cb +
      wave_data[[paste0(region, "_hot_wave3")]] +
      wave_data[[paste0(region, "_cold_wave3")]],
    family = gaussian()
  )
  crosspred(cb, fit, cen = overall_centers[[region]][[age_group]],
            at = -5:40, model.link = "log")
}

plot_overall_region <- function(region) {
  predictions <- setNames(lapply(age_names, function(age_group) {
    fit_overall_pred(region, age_group)
  }), age_names)

  pdf(file.path(figure_overall_dir, paste0(region, ".pdf")), width = 10, height = 6.18)
  par(mfrow = c(2, 2), mar = c(5, 5, 2, 1), oma = c(1, 1, 3, 1),
      cex.lab = 2, cex.axis = 2, cex.main = 2)
  for (age_group in age_names) {
    plot(
      predictions[[age_group]], "overall",
      xlab = "UTCI", ylab = "RR", xlim = c(-5, 40),
      ylim = overall_ylim[[region]][[age_group]],
      ci.arg = list(lty = 2, lwd = 2, col = "#FFBE7A"),
      main = age_titles[[age_group]]
    )
  }
  dev.off()
  predictions
}
```

```r
overall_predictions <- setNames(lapply(regions, plot_overall_region), regions)
```

```r
knitr::include_graphics(file.path(figure_overall_dir,
                                  c("Athens.pdf", "Lisbon.pdf", "Rome.pdf")))
```





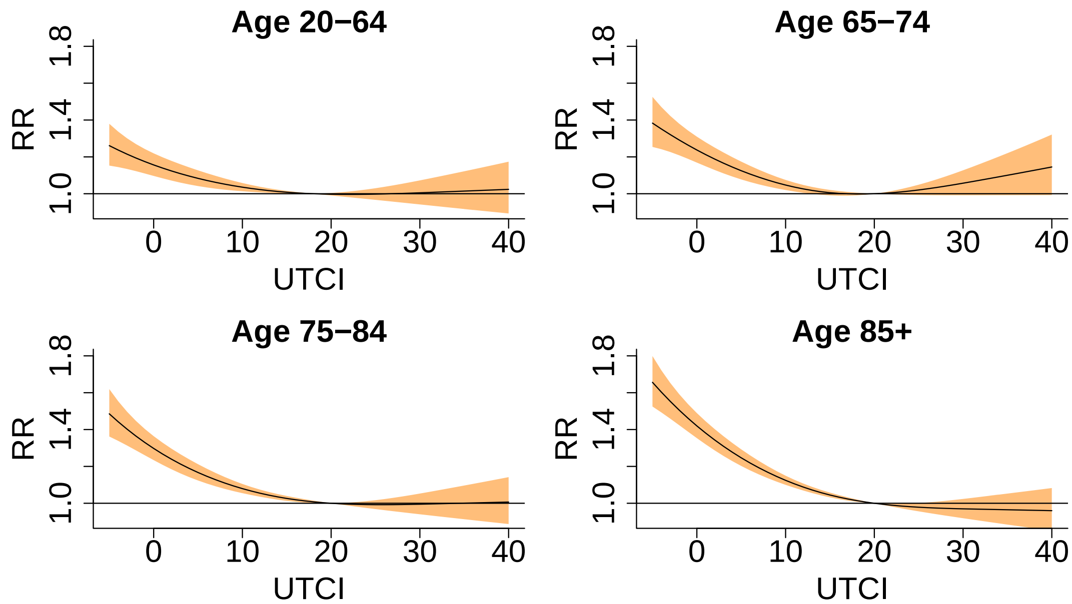


Table 1. HWD and CWD coefficients

```r
significance_stars <- function(p_value) {
  ifelse(p_value < 0.01, "^{***}",
         ifelse(p_value < 0.05, "^{**}",
                ifelse(p_value < 0.1, "^{*}", "")))
}

format_beta <- function(estimate, p_value) {
  paste0("$", formatC(estimate, format = "f", digits = 4),
         significance_stars(p_value), "$")
}

fit_wave_model <- function(temp_data, region, age_group) {
  cb <- cb_list[[region]]
  wave_data <- wave_list[[region]]
  y <- temp_data[[region]][[age_group]]
  glm(
    y ~ cb +
      wave_data[[paste0(region, "_hot_wave3")]] +
      wave_data[[paste0(region, "_cold_wave3")]],
    family = gaussian()
  )
}

extract_wave_terms <- function(model) {
  coef_table <- summary(model)$coefficients
  p_col <- grep("^Pr\\(", colnames(coef_table), value = TRUE)
  terms <- c(
    HWD = grep("hot_wave3", rownames(coef_table), value = TRUE),
    CWD = grep("cold_wave3", rownames(coef_table), value = TRUE)
```

```r
  )
  data.frame(
    term = names(terms),
    estimate = unname(coef_table[terms, "Estimate"]),
    p_value = unname(coef_table[terms, p_col]),
    formatted = format_beta(unname(coef_table[terms, "Estimate"]),
                            unname(coef_table[terms, p_col])),
    row.names = NULL
  )
}

collect_wave_coef <- function(temp_data, model_label) {
  rows <- list()
  for (age_group in age_names) {
    for (region in regions) {
      wave_terms <- extract_wave_terms(fit_wave_model(temp_data, region, age_group))
      rows[[length(rows) + 1]] <- data.frame(
        age_group = age_labels[[age_group]],
        model = model_label,
        region = region,
        wave_terms,
        row.names = NULL
      )
    }
  }
  do.call(rbind, rows)
}

make_wave_coef_latex <- function(coef_table) {
  get_cell <- function(age, model, region, term) {
    cell <- coef_table[
      coef_table$age_group == age & coef_table$model == model &
        coef_table$region == region & coef_table$term == term,
      "formatted"
    ]
    if (length(cell) == 0) return("")
    cell[[1]]
  }

  lines <- c(
    "\\begin{table}[H]",
    "\\centering",
    "\\renewcommand{\\arraystretch}{1.2}",
    "\\resizebox{0.95\\textwidth}{!}{%",
    "\\begin{tabular}{cllllll}",
    "    \\toprule",
    "    \\multirow{2}{*}{\\textbf{Age group}}",
    "      & \\multirow{2}{*}{\\textbf{Model}}",
    "      & \\multicolumn{2}{c}{\\textbf{Athens}}",
    "      & \\multicolumn{2}{c}{\\textbf{Lisbon}}",
    "      & \\multicolumn{2}{c}{\\textbf{Rome}} \\\\",
    "    \\cline{3-8}",
    "      &",
```

```
    "      & $\\beta_{\\text{HWD}_t}$ & $\\beta_{\\text{CWD}_t}$",
    "      & $\\beta_{\\text{HWD}_t}$ & $\\beta_{\\text{CWD}_t}$",
    "      & $\\beta_{\\text{HWD}_t}$ & $\\beta_{\\text{CWD}_t}$ \\\\",
    "    \\midrule"
  )

  for (age_index in seq_along(age_labels)) {
    age <- age_labels[[age_index]]
    for (model_index in seq_along(c("DLNM--LC", "DLNM--LL"))) {
      model <- c("DLNM--LC", "DLNM--LL")[[model_index]]
      age_cell <- ifelse(model_index == 1, age, "")
      lines <- c(lines, paste0(
        "    ", age_cell, " & ", model, " & ",
        get_cell(age, model, "Athens", "HWD"), " & ",
        get_cell(age, model, "Athens", "CWD"), " & ",
        get_cell(age, model, "Lisbon", "HWD"), " & ",
        get_cell(age, model, "Lisbon", "CWD"), " & ",
        get_cell(age, model, "Rome", "HWD"), " & ",
        get_cell(age, model, "Rome", "CWD"), " \\\\"
      ))
    }
    if (age_index < length(age_labels)) lines <- c(lines, "    \\midrule")
  }

  paste(c(
    lines,
    "\\bottomrule",
    "\\noalign{\\vskip 6pt}",
    "\\end{tabular}",
    "}",
    "\\footnotesize{Level of significance: $0.01$:***, $0.05$:**, $0.1$:*}",
    "\\caption{\\small Coefficients of $\\text{HWD}_t$ and $\\text{CWD}_t$ in the DLNM component of DLNI
    "\\label{hwd_cwd_coefficients_combined}",
    "\\end{table}"
  ), collapse = "\n")
}

DLNM_wave_coef_LC <- collect_wave_coef(LC_temp, "DLNM--LC")
DLNM_wave_coef_LL <- collect_wave_coef(LL_temp, "DLNM--LL")
DLNM_wave_coef_combined <- rbind(DLNM_wave_coef_LC, DLNM_wave_coef_LL)
DLNM_wave_coef_latex <- make_wave_coef_latex(DLNM_wave_coef_combined)

saveRDS(DLNM_wave_coef_LC, file.path(results_dir, "DLNM_wave_coef_LC.rds"))
saveRDS(DLNM_wave_coef_LL, file.path(results_dir, "DLNM_wave_coef_LL.rds"))
saveRDS(DLNM_wave_coef_combined,
        file.path(results_dir, "DLNM_wave_coef_combined.rds"))
writeLines(DLNM_wave_coef_latex,
           file.path(results_dir, "DLNM_wave_coef_combined_table.tex"))

cat(DLNM_wave_coef_latex)
```

Age group	Model	Athens		Lisbon		Rome	
		β_{HWD_t}	β_{CWD_t}	β_{HWD_t}	β_{CWD_t}	β_{HWD_t}	β_{CWD_t}
20–64	DLNM–LC	0.0132***	0.0067	0.0138**	0.0100**	0.0038	0.0077**
	DLNM–LL	0.0132***	0.0069	0.0140**	0.0104***	0.0037	0.0088**
65–74	DLNM–LC	0.0095***	0.0097**	0.0205***	0.0039	−0.0112**	0.0015
	DLNM–LL	0.0094***	0.0107***	0.0206***	0.0039	−0.0117***	0.0010
75–84	DLNM–LC	0.0161***	0.0060*	0.0181***	0.0125***	0.0089**	0.0066**
	DLNM–LL	0.0161***	0.0053	0.0182***	0.0121***	0.0081**	0.0052
85+	DLNM–LC	0.0198***	0.0007	0.0198***	0.0134***	0.0184***	0.0080**
	DLNM–LL	0.0197***	0.0004	0.0199***	0.0130***	0.0177***	0.0070**

Level of significance: 0.01:***, 0.05:**, 0.1:*

Table 1: Coefficients of HWD_t and CWD_t in the DLNM component of DLNM–LC and DLNM–LL models.

Figure 6. Climate loading theta

```r
theta_values <- unlist(c(theta_LC, theta_LL))
min_val <- min(theta_values, na.rm = TRUE) -
  0.05 * abs(min(theta_values, na.rm = TRUE))
max_val <- max(theta_values, na.rm = TRUE) +
  0.05 * abs(max(theta_values, na.rm = TRUE))
step_size <- (max_val - min_val) / 4

plot_theta <- function(region, age_group) {
  df <- data.frame(
    time = seq(as.Date("2015-01-01"), as.Date("2019-12-30"), length.out = 260),
    DLNM_LC = as.numeric(theta_LC[[region]][[age_group]]),
    DLNM_LL = as.numeric(theta_LL[[region]][[age_group]])
  )
  df_long <- pivot_longer(df, cols = c("DLNM_LC", "DLNM_LL"),
                          names_to = "Model", values_to = "Value")
  df_long$Model <- factor(df_long$Model, levels = c("DLNM_LC", "DLNM_LL"))

  ggplot(df_long, aes(x = time, y = Value, color = Model)) +
    geom_line(size = 0.3) +
    scale_color_manual(
      name = "", values = c("DLNM_LC" = "black", "DLNM_LL" = "#008F7A"),
      labels = c("DLNM-LC", "DLNM-LL")
    ) +
    scale_x_date(limits = as.Date(c("2015-01-01", "2020-01-01")),
                 date_breaks = "1 year", date_labels = "%Y", expand = c(0, 0.01)) +
    scale_y_continuous(limits = c(min_val, max_val),
                       breaks = seq(min_val, max_val, by = step_size),
                       labels = number_format(accuracy = 0.01)) +
    labs(title = paste0(age_titles[[age_group]], " (", region, ")"),
         x = "Year", y = expression(theta)) +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5, size = 7, face = "plain"),
```

```r
      panel.grid = element_blank(),
      panel.border = element_rect(color = "black", fill = NA, size = 1),
      legend.position = "none"
    )
}

theta_plots <- list()
for (region in regions) {
  for (age_group in age_names) {
    theta_plots[[paste(region, age_group)]] <- plot_theta(region, age_group)
  }
}

theta_plot <- (theta_plots[["Athens Y20_64"]] | theta_plots[["Athens Y65_74"]] |
                 theta_plots[["Athens Y75_84"]] | theta_plots[["Athens Y_GE85"]]) /
  (theta_plots[["Lisbon Y20_64"]] | theta_plots[["Lisbon Y65_74"]] |
     theta_plots[["Lisbon Y75_84"]] | theta_plots[["Lisbon Y_GE85"]]) /
  (theta_plots[["Rome Y20_64"]] | theta_plots[["Rome Y65_74"]] |
     theta_plots[["Rome Y75_84"]] | theta_plots[["Rome Y_GE85"]]) +
  plot_layout(guides = "collect") &
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
        legend.position = "bottom")

ggsave(file.path(figure_calibration_dir, "fraction.pdf"),
       theta_plot, width = 10, height = 6.18)
print(theta_plot)
```